SEGi UNIVERSITY

FACULTY OF ENGINEERING, BUILT ENVIRONMENT & INFORMATION TECHNOLOGY

Bachelor of Information Technology (Hons)
Bachelor of Computer Science (Hons) in Cybersecurity

Final Year Project (FYP) Handbook

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IMPORTANT

Plagiarism is a serious offence.

Copy and paste of report content is prohibited.

1. Introduction

All final year computing program students are required to undertake Final Year Project (FYP) prior to graduation. FYP is divided into two parts: FYP1 and FYP2 and are run in two semesters respectively.

Students are required to conduct research, design, and develop software systems or applications on real-world problems to produce practical solutions. It is an opportunity for students to use various problem-solving tools and techniques to solve problems encountered throughout the activities. Students' learning process is gained through hands on experience. Students are also required to apply project management skills in managing the project. Students will be working under the close supervision of the supervisors assigned to them, to ensure that they are working on the projects according to guidelines provided and to complete the project on time. The supervisors will guide the students to follow the project direction to achieve the project objectives.

2. Final Year Project Objectives

a. The objectives for FYP1:
   i. Identifying suitable and relevant topic which can be developed either through development or research activities that will match the level expected of an undergraduate student.
   ii. Identifying relevant information pertaining to project's requirements from a variety of resources.
   iii. Explaining and applying project development or research methodologies appropriate to the project.
   iv. Plan, design, and propose a feasible project based on the given timeline.
   v. Communicating project ideas and final product through technical report and presentation.

b. The objectives of FYP2:
   i. Develop the project proposed in FYP1 according to the proposed plan and design.
   ii. Verify and validate the developed projects against the proposed objectives goals.
   iii. Proposing future improvement based on project outcomes.
   iv. Communicating project ideas and final product through technical report and presentation.

2.1 Scope of Work

The project scope includes a workable software product, project final report, and presentation. Students should use available CASE tools to design the system, decide an appropriate programming language to develop the system, and finally write a report to communicate project ideas and final product and to document the process of product development. The report must be written with clear sentences, correct grammar and spelling, as well as following the format provided. The area and scope of the project should be narrowed down so that the project is feasible and couldbe completed within the allocated time frame. The project work must exhibit an element of originality, which indicates the maturity level of a final year student.

2.2 Final Year Project Flow

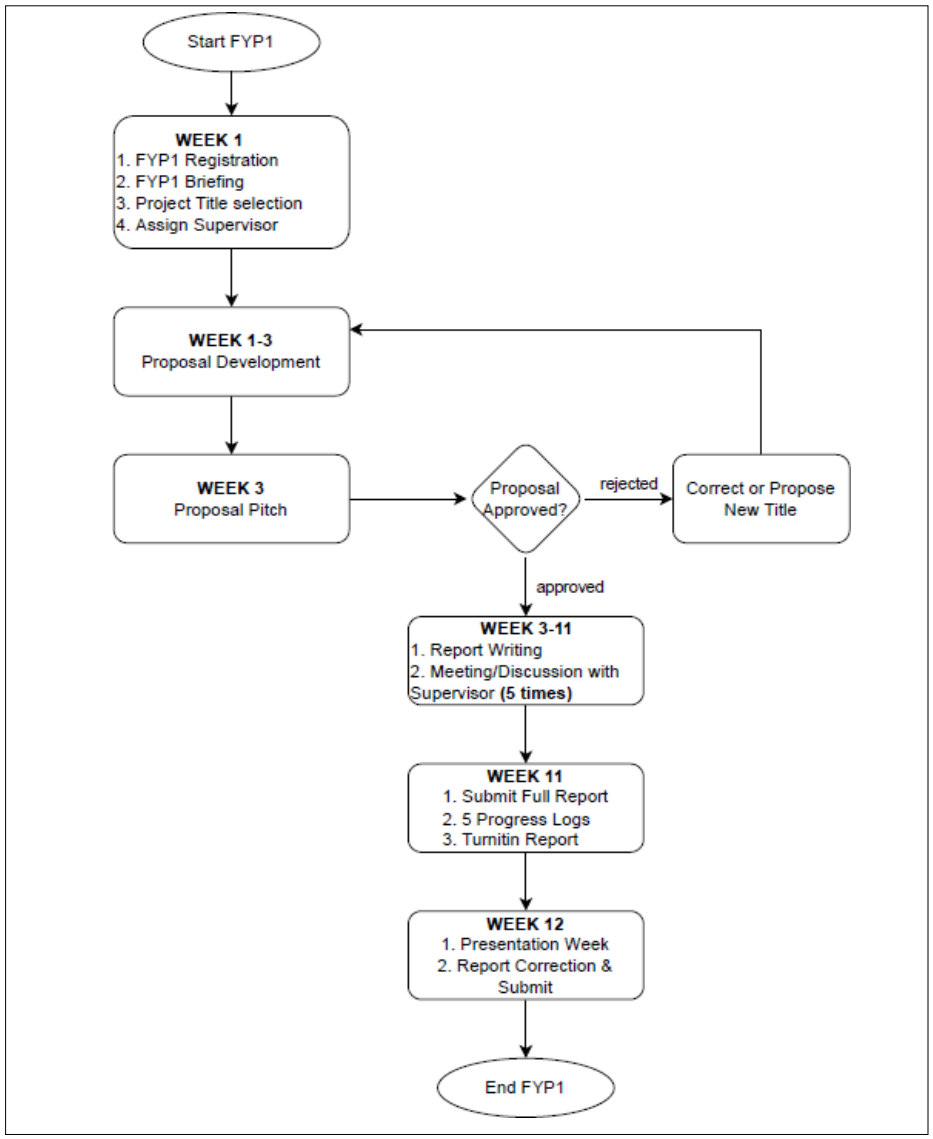


Figure 1: FYP Process Flow

3. Project Deadlines

FYP deadlines are provided in Table 1. However, students are encouraged to meet their supervisors AT LEAST FIVE (5) times throughout the semester.

Table 1: FYP Deadlines

Tasks	Deadlines
Submission of Project Proposal Form (PPF)	Week 3
Submission of Final Year Project Report	Week 11
FYP Presentation (FYP1)/System Demo (FYP2)	Week 11-12

4. Responsibilities

4.1 Students

Students are required to demonstrate their ability to integrate fundamental knowledge in developing techniques, methods, and analyses. Students should take their own initiatives such as to confirm their project title and to work independently through self-discipline, self-management and job coordination while undertaking the project.

Students' responsibilities are:

a) To identify a suitable research project area. The project topic can be of their own or suggested by the supervisor. The proposed project must be related to the student's programme discipline and specialisation.
b) To plan meetings with their supervisor at least FIVE (5) times throughout the semester. Progress must be shown during the meeting.
c) To adhere to the meeting schedule with the supervisor for the purpose of updating the progress and seeking advice on project.
d) To be responsible in finding alternative solutions for problems encountered such as computer crashes and instrument failure.
e) To follow FYP report format and guidelines provided.

4.2 Supervisor

Students will be supervised by one main supervisor, who is knowledgeable in the relevant field of expertise.

Supervisors' responsibilities are:

a) To provide guidance on the selection of project topics acceptable for undergraduate final year projects.

b) To assist and guide students on the project and standard expected, techniques, methods used for the project and finally to provide feedback to the students on time.

c) To meet regularly (biweekly) to ensure the progress made and verify them by signing the Progress LogForm (PLF).

d) To ensure students submit draft report before the final submission and to provide constructive feedback promptly.

4.2.1 Moderator

Students will be examined by the moderators. The moderators are responsible to attend the oral presentation and assess the students report based on the assessment rubrics. **APPENDIX A** – Marking Rubric

5. Report Submission

Each student is required to submit the project report by the deadline provided. The project submission should include (Table 2):

Table 2: List of FYP submission

FYP1	FYP2
- FYP Report (Chapter 1 – 3) - Project Proposal Form - Progress Log Form - Turnitin Report (<20%)	- FYP Report (Chapter 1 – 6) - Project Proposal Form - Progress Log Form (FYP2 only) - Turnitin Report (<20% - for Chapter 4-6 only) - Working System (Product)

Late submission is not acceptable and will be rejected and student are not allowed to present. Other criteria for failing FYP are:

- No regular meeting with supervisor and/or
- Incomplete/low quality Report and/or
- System not working and/or
- Absent for oral presentation/system demo

6. FYP Report Content

Standard chapter sizes in number of pages are shown below. Exceptionally, individuals may wish to negotiate these numbers with their supervisor. The lengths of chapters below are a VERY ROUGH GUIDLINE. The standard report format is (in strict order):

i.	Cover Page	
ii.	Acknowledgement (Optional)	
iii.	Abstract	
iv.	Table of Contents	
v.	List of Figures	
vi.	List of Tables	
vii.	List of Appendices	
Chapter 1	Introduction	5 – 10 pages
Chapter 2	Literature Review	10 – 15 pages
Chapter 3	Methodology	15 – 25 pages
Chapter 4	System Implementation	15 – 25 pages
Chapter 5	Results	10 – 15 pages
Chapter 6	Conclusion & Future Works	5 -10 pages

7. Assessment

Students are assessed based on their capability in undertaking the project and their general effort in fulfilling the specific learning outcomes set by the university. The components of assessment and the grading structure are shown in Table 3:

Table 3: Assessment structure

FYP	Components	(%)	Assess by
FYP1	Project Proposal	10	Supervisor (70%) Moderator (30%)
	Presentation/Supervisor	30	
	FYP1 Report (Chapter 1 – 3)	60	
FYP2	Product Demo/Supervisor	50	Supervisor (70%) Moderator (30%)
	Report Format	10	
	FYP2 Report (Chapter 4 – 6)	40	

8. Project Area

Table 4 shows list of project areas to be considered.

Table 4: Project Area

Business Systems Design (BSD)	Software Engineering (SE)	Computer Network (CN)
• Information Systems • Information Systems Development • Decision Support Systems • Project Management • Database • Information Retrieval • Decision Support System (DSS) • Customer Relationship Management (CRM) • Knowledge Management System Development	• Soft computing • Software System Development Evolution • Software Design/ Architecture • Software Quality • Software Modelling and Process • Software Testing • Software Requirement • Software Reusability • Embedded Real-time Software, • Requirement Traceability • Artificial Intelligence (Neural network, Genetic Algorithm, Support Vector Machine, Fuzzy Logic, Speech Processing) • Planning and Scheduling • Automata and Formal Language • Business Rules and Design Pattern • Web-based Application • Human Computer Interaction • Expert System • Image Processing and Pattern Recognition • Computer Games • Visualization\Object Modeling • Virtual Environment - Artificial Intelligence • Web Technology	• Wi-Fi Network Manager • Network Monitoring System Based on OpenNMS • Bluetooth Software Controller • Bus Speed Monitoring System Using GPS • Network Simulation Using OPNET

9. Plagiarism

Plagiarism is the act of obtaining or attempting to obtain credit for academic work by representing the work of another as one's own without the necessary and appropriate acknowledgement. Student should submit the Turnitin Report along with the FYP report. The report is considered as plagiarised work if the percentage of similarity is more than 20%. If a student is caught committing plagiarism, the student is required to retake the module.

10. Barring

Students shall be barred from taking the oral presentation and/or system demo for the following reasons:

a) Fail to submit the final report by the deadline.

b) No regular meetings with supervisor / no progress shown to the supervisor throughout the semester.

Consequently, the student's performance is deemed to be disqualified and a grade "F" shall be given to this course by the examiners.

11. Copyright

The university shall be the owner for all findings, designs, patents, and other intellectual property rights.